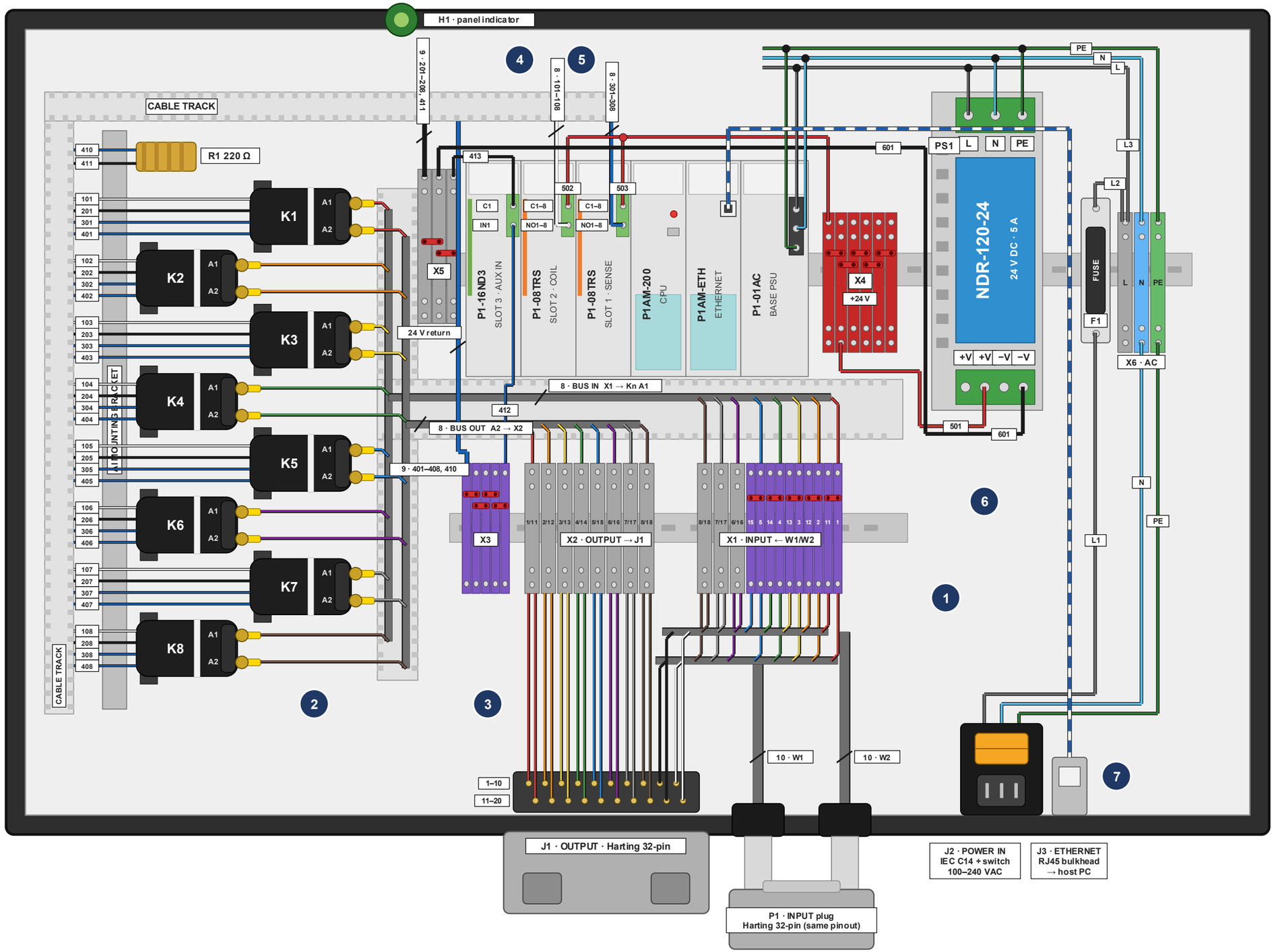
3rd Order Shim PLC Switch Box Wiring Diagram



WHAT THIS BOX DOES

The box connects or disconnects 8 power channels between an input cable and an output connector using 8 high-power relays (contactors K1–K8). A programmable logic controller (PLC) switches each contactor on command from a host computer and confirms that it switched.

HOW IT WORKS

1. **Input.** The input cable (W1 + W2, 20 wires) lands on terminal block X1. Each pin is paired with the pin ten higher (1 & 11, 2 & 12 …) to form one channel.
2. **Switch.** Each channel passes through one contactor, K1–K8. Closed = connected, open = disconnected.
3. **Output.** Terminal block X2 splits each channel back onto its two pins at output connector J1, matching the input pinout.
4. **Command.** A relay in the PLC (slot 2) sends 24 V to a contactor's coil to close it.
5. **Check.** A second PLC relay (slot 1) briefly sends 24 V through the auxiliary contact on each contactor, one at a time. A PLC input (slot 3) reads the result to confirm each contactor is in the commanded open or closed state.
6. **Power.** Building AC enters at J2 through fuse F1. PS1 converts it to 24 V DC for the controls; the PLC has its own supply.
7. **Network.** A host computer sends open / close commands over Ethernet (J3).

WIRE COLORS

Power channels 1–8 (each color = one channel, both pins):

1 · 2 · 3 · 4 · 5 · 6 · 7 · 8

Pins 9/19 and 10/20: straight through, not switched

- Power to relay coil
- Auxiliary contact
- +24 V DC control power
- 24 V return (0 V)
- Mains live (fused)
- Mains neutral
- Earth (safety ground)
- Ethernet (dashed)

SYMBOLS

- Cable or bundle; the number next to the slash = wires inside
- Red jumper linking neighbouring terminals
- 203 — Wire number: 1xx coil, 2xx coil return, 3xx/4xx auxiliary contact, 5xx +24 V, 6xx 24 V return
- Dot = joined. Wires that cross without a dot are not connected
- Terminal blocks: purple = 1 wire each end, grey = 2 wires each end

MAIN PARTS

Ref	What it is	Part
K1–K8	50 A / 1000 V DC contactor	Littelfuse DCNHE50MF24-F
PLC	Controller + I/O cards	AutomationDirect P1AM-200
PS1	AC → 24 V DC power supply	Mean Well NDR-120-24
J1 / P1	Output socket / input plug, 32-pin	Harting 32-pin connector

H1 · panel indicator

CABLE TRACK

9 · 201–208, 411

8 · 101–108

8 · 301–308

410
411
R1 220 Ω

101 201 301 401 — K1 (A1 / A2)
102 202 302 402 — K2 (A1 / A2)
103 203 303 403 — K3 (A1 / A2)
104 204 304 404 — K4 (A1 / A2)
105 205 305 405 — K5 (A1 / A2)
106 206 306 406 — K6 (A1 / A2)
107 207 307 407 — K7 (A1 / A2)
108 208 308 408 — K8 (A1 / A2)

A1 MOUNTING BRACKET

CABLE TRACK

413

502

503

X5

24 V return

P1-16ND3 · SLOT 3 · AUX IN
C1
IN1

P1-08TRS · SLOT 2 · COIL
C1–8
NO1–8

P1-08TRS · SLOT 1 · SENSE
C1–8
NO1–8

P1AM-200 · CPU

P1AM-ETH · ETHERNET

P1-01AC · BASE PSU

601

X4
+24 V

PS1
L N PE
NDR-120-24
24 V DC · 5 A
+V +V −V −V

501

601

PE
N
L

L3
L2
L1

FUSE
F1

L N PE
X6 · AC

8 · BUS IN X1 → Kn A1

412

8 · BUS OUT A2 → X2

9 · 401–408, 410

X3

X2 · OUTPUT → J1
1/11 2/12 3/13 4/14 5/15 6/16 7/17 8/18

X1 · INPUT ← W1/W2
8/18 7/17 6/16 15 5 14 4 13 3 12 2 11 1

1–10
11–20

10 · W1

10 · W2

N

PE

L1

J1 · OUTPUT · Harting 32-pin

P1 · INPUT plug
Harting 32-pin (same pinout)

J2 · POWER IN
IEC C14 + switch
100–240 VAC

J3 · ETHERNET
RJ45 bulkhead
→ host PC